

From Rate Suppression to Outcome Starvation: A Finite-Width Causal Theory of Recurrent Learning

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Abstract

A strong predictive cue can accelerate optimization while preventing a weaker but independently predictive cue from being learned. We study this gradient-starvation question in recurrent binary classifiers using a paired causal intervention: a both-feature trajectory is compared with a same-initialization trajectory in which only the strong input channel is ablated. Our contribution is a finite-width package rather than a claimed general dynamical mean-field theory (DMFT). First, we derive exact finite-width signed-logit and response dynamics through the full sample empirical NTK and response-logit cross-kernel. Freezing these objects at initialization defines a nonlinear logistic tangent surrogate; continuous Grönwall and experiment-matched discrete secant-kernel bounds compare it with the trained model conditionally on kernel movement. Second, after a transfer-to-suppression rate crossing, we prove that outcome starvation occurs if and only if the negative post-crossing drift area exhausts the earlier transfer advantage. Third, for a noiseless rank-one cue law, an ordering-invariant log drift ratio gives sufficient conditions for a unique rate crossing. We also prove transverse hitting-time stability under quantitative first-entry hypotheses and an exact six-scalar, zero-disorder/zero-lag wide limit with an explicit $9/(N\epsilon^2)$ initialization bound. Finally, we derive Counterfactual Drift Correction (CDC), the unique uncapped minimum Euclidean-norm correction in the fixed implemented parameter coordinates that matches a weak-only drift target while preserving instantaneous strong-response drift, and prove a local $O(\eta^2)$ one-step deviation bound in that same chart.

Controlled dense-linear experiments exhibit positive weak-learning AUC gaps in 32/32 seeds at lags 0, 2, and 4. Under the final direct-autograd nonlinear protocol, tanh transfer-to-starvation is seed-dependent (3/8 causal certificates), while GRU has 5/8 response crossings but 0/8 causal certificates because the weak-only model never passes the learnability gate. A sealed five-endpoint frozen- kernel pilot fails its preregistered rule on tanh learnability. After that failure, a fresh restricted 4×8 data/model-seed factorial classifies response- crossing events in 64/64 architecture-seed records and drift-crossing events in 60/64; two-way-bootstrap 95% lower endpoints are 1.0 and 0.8125. These are event- classification results at one controlled cell, not validated phase, causal, learnability, time, magnitude, or kernel-stability claims. CDC uniquely preserves instantaneous strong drift, but does not improve weak rescue or final strong retention over all shadow-based ablations. A general positive-disorder/positive- lag optimization-time DMFT, an analytic phase boundary, and certified or quantitatively accurate nonlinear crossover times remain open.

1 Introduction

Cross-entropy training can favor a subset of predictive features strongly enough that other useful features are learned slowly or not at all. This phenomenon was formalized as gradient starvation by Pezeshki et al. [2021]. Recurrent networks add a temporal geometry: cue strength, lag, shared recurrent parameters, and evolving feature response geometry all affect which signal receives gradient. The resulting mechanism is easy to overstate. A weak-feature drift can decline without changing sign; its relative drift can change sign without the weak response falling behind; and a response gap is not evidence of causal starvation if the weak-only counterfactual is itself unable to learn.

We address these distinctions with a paired intervention. The both-feature and weak-only models share initialization, labels, samples, weak coordinates, and noise; only the strong input channel is set to zero in the weak-only condition. We track common feature-response functionals in both trajectories. This setup separates three statements:

1. *relative transfer*: the strong cue initially increases the weak response rate;
2. *rate suppression*: the both-feature weak response later grows more slowly than its counterfactual;
3. *causal outcome starvation*: the both-feature weak response falls behind and the weak-only counterfactual demonstrably learns.

The distinction is both mathematical and empirical: the final GRU experiment has response crossings but fails the causal learnability gate in every seed.

Contributions.

1. We prove an exact finite-width response-flow identity for logistic gradient flow. A closed projected equation $\dot{M} = Gg$ follows only under exact, parameter-independent logit realization; nonlinear probe responses carry an explicit residual.
2. We prove exact finite-width signed-logit and response dynamics through the full sample empirical NTK and response-logit cross-kernel. Freezing them gives an initialization-only nonlinear logistic surrogate; conditional continuous and exact secant-SGD bounds transfer supplied kernel-movement control into response, sign, crossing, and hitting-time control.
3. We give a necessary-and-sufficient negative-tail-area criterion converting a unique rate crossover into an outcome crossing, together with quantitative crossing-time bounds.
4. For noiseless rank-one cues, we derive an ordering-invariant log drift ratio and sufficient conditions for a unique rate crossover. Unlike a non-canonical additive geometry/field split, the ratio has a fixed meaning.
5. We prove stability of quantitatively isolated transverse first hitting times and an exact zero-disorder, zero-lag six-scalar closure with finite-horizon convergence in probability and explicit deterministic tube constants.
6. In the fixed implemented Euclidean/Frobenius parameter chart, we prove three local CDC results: exact instantaneous strong-drift preservation, unique uncapped minimum Euclidean-norm optimality when feasible, and a local $O(\eta^2)$ one-step strong-response deviation bound.
7. We provide executable theorem contracts and a claim-audited experiment suite. The corrected nonlinear evidence is reported as seed-dependent and the unsuccessful exploratory search is retained as a negative result.

Claim boundary. We do not claim that recurrent networks universally undergo transfer-to-starvation, that a general joint CE-RNN optimization-time DMFT has been solved, or that an analytic ρ_c is known. The frozen empirical-NTK surrogate is exact for its initialization-linearized tangent model, not for trained tanh/GRU networks: no required instantaneous or secant-kernel stability bound is proved here. The held-out factorial validates only binary drift/response-crossing classification at one width/task cell; it does not validate phase, causal starvation, weak-only learnability, calibrated trajectories, crossing times, or prevalence. The general DMFT solver remains deliberately unimplemented where its proof obligations are unmet.

2 Related work

Gradient starvation. Pezeshki et al. [2021] identify feature starvation under cross-entropy and motivate Spectral Decoupling. Our focus is different: recurrent temporal features, a paired strong-channel intervention, exact finite-width response dynamics, and the logical gap between rate suppression and outcome starvation.

Learning dynamics in recurrent networks. Ger and Barak [2026] derive low-dimensional overlap learning dynamics for low-rank RNNs and show that overlaps invisible to the loss can retain training history. This directly cautions against closing a recurrent learning theory using only visible response coordinates. Clark et al. [2026] develop a mean-field account of structure and disorder in task-trained recurrent circuits in a long-time Langevin/Gibbs setting. That equilibrium construction is distinct from the paired, noiseless

optimization-time gradient flow studied here. Accordingly, we claim neither the first learning theory for RNNs nor a completed general RNN DMFT; our novelty lies in the causal pairing and finite-width rate-to-outcome results.¹

Gradient surgery. PCGrad projects conflicting task gradients [Yu et al., 2020]; Bloop stabilizes an auxiliary projection using an exponential moving average [Hsieh et al., 2024]. Projection itself is not our novelty. CDC protects the gradient of a theory-defined strong-feature response and takes its target from a matched causal weak-only shadow. This stronger information setting must be reported: CDC requires a constructible ablation, a second model, lockstep training, and full-batch gradients.

3 Causal setup and definitions

3.1 Paired logistic gradient flow

For condition $C \in \{\text{B}, \text{W}\}$, let $\theta_C(\tau) \in \mathbb{R}^P$ follow full-batch gradient flow on signed logits $r_i^C(\theta) = y_i f_C(x_i; \theta)$:

$$L_C(\theta) = \frac{1}{n_C} \sum_{i=1}^{n_C} \log(1 + e^{-r_i^C(\theta)}), \quad \dot{\theta}_C = -\nabla L_C(\theta_C). \quad (1)$$

The two conditions share initialization and all data except that the strong channel is zeroed under W. Let M_s^C, M_w^C be common differentiable strong and weak response functionals. Shared initialization gives

$$\Delta(0) = 0, \quad \Delta(\tau) = M_w^{\text{B}}(\tau) - M_w^{\text{W}}(\tau). \quad (2)$$

Optimization time τ is distinct from sequence index k .

The controlled dense linear RNN is

$$h_{k+1} = Wh_k + Bx_k, \quad f(x) = c^\top h_T, \quad (3)$$

with $W_{ij}(0) \sim \mathcal{N}(0, g^2/N)$, $B_{ia}(0) \sim \mathcal{N}(0, 1/N)$, and $c_i(0) \sim \mathcal{N}(0, 1/N)$. The width scaling of B is essential for a non-divergent projected metric. We additionally study tanh RNNs and GRUs. For nonlinear models the two responses are common odd symmetric unit-probe contrasts, $[f(x_a^+) - f(x_a^-)]/2$, not assumed additive coordinates of every data logit.

3.2 Causal labels and metrics

Let

$$d(\tau) = \Delta'(\tau) = F_w^{\text{B}}(\tau) - F_w^{\text{W}}(\tau). \quad (4)$$

We call $d > 0$ relative transfer and $d < 0$ relative rate suppression. An outcome crossing occurs when Δ reaches zero after a positive excursion and then becomes negative. We reserve *causal starvation* for outcome suppression when the weak-only trajectory also reaches a preregistered target β within horizon H . If the weak-only model does not reach β , the result is indeterminate rather than starvation.

We report first-hitting-time delay when both hits are observed. The horizon-wide weak AUC gap is

$$\mathcal{A}_w = \int_0^H (M_w^{\text{W}}(\tau) - M_w^{\text{B}}(\tau)) \, d\tau, \quad (5)$$

computed by the logged-grid trapezoid rule. GSI-5 is retained as a margin-weight diagnostic, not as the definition of starvation.

¹The descriptions of the 2026 sources were paraphrased from the linked public sources. Content was rephrased for compliance with licensing restrictions.

4 Exact finite-width response dynamics

Theorem 1 (Universal response-logit cross-kernel flow). *At every finite width and every optimization time at which M_a^C is differentiable,*

$$\dot{M}_a^C = \frac{1}{n_C} \sum_i K_{ai}^C \sigma(-r_i^C), \quad K_{ai}^C = \langle \nabla M_a^C, \nabla r_i^C \rangle. \quad (6)$$

No width limit or two-mode realization is required.

Corollary 2 (Closed projected flow under exact realization). *If the signed logits admit the exact parameter-independent representation $r_i^C = \sum_b z_{ib}^C M_b^C$, then*

$$\dot{M}_a^C = \sum_b G_{ab}^C g_b^C, \quad G_{ab}^C = \langle \nabla M_a^C, \nabla M_b^C \rangle, \quad g_b^C = \frac{1}{n_C} \sum_i z_{ib}^C \sigma(-z_i^C \cdot M^C). \quad (7)$$

If $r_i^C = z_i^C \cdot M^C + r_i^\perp$, the exact equation instead contains

$$R_a^C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \langle \nabla M_a^C, \nabla r_i^\perp \rangle, \quad \dot{M}_a^C = (G^C g^C)_a + R_a^C. \quad (8)$$

Theorem 3 (Exact full empirical-NTK dynamics). *Let $J^C(\tau) \in \mathbb{R}^{n_C \times P}$ have row $\nabla r_i^C(\theta_C(\tau))^\top$, and define*

$$\Theta^C(\tau) = J^C(\tau) J^C(\tau)^\top, \quad c_a^C(\tau) = J^C(\tau) \nabla M_a^C(\theta_C(\tau)). \quad (9)$$

Writing $q(r) = \sigma(-r)$ componentwise, the exact finite-width flow is

$$\dot{r}^C = \frac{1}{n_C} \Theta^C(\tau) q(r^C), \quad \dot{M}_a^C = \frac{1}{n_C} c_a^C(\tau)^\top q(r^C). \quad (10)$$

No width limit or response-coordinate realization is required.

Definition 4 (Initialization-frozen nonlinear logistic surrogate). *For one condition, let $\Theta_0 = \Theta(0)$ and $c_0 = c_a(0)$. Define*

$$\dot{\bar{r}} = \frac{1}{n} \Theta_0 q(\bar{r}), \quad \dot{\bar{M}} = \frac{1}{n} c_0^\top q(\bar{r}), \quad (\bar{r}(0), \bar{M}(0)) = (r(0), M(0)). \quad (11)$$

The gate remains nonlinear. This system is exact for the initialization-linearized tangent model, not for the trained nonlinear network.

Theorem 5 (Conditional continuous and secant-SGD comparison). *Use Euclidean and induced operator norms. Suppose on $[0, t]$ that $\|\Theta(u) - \Theta_0\| \leq B_\Theta$, $\|c(u) - c_0\| \leq B_c$, and both logistic weight vectors have norm at most V . Set $a = \|\Theta_0\|/(4n)$,*

$$\phi_a(t) = \frac{e^{at} - 1}{a}, \quad \psi_a(t) = \frac{e^{at} - 1 - at}{a^2}, \quad (12)$$

with $\phi_0(t) = t$ and $\psi_0(t) = t^2/2$. Then

$$\|r(t) - \bar{r}(t)\| \leq \frac{B_\Theta V}{n} \phi_a(t), \quad (13)$$

$$|M(t) - \bar{M}(t)| \leq \frac{B_c V}{n} t + \frac{\|c_0\| B_\Theta V}{4n^2} \psi_a(t). \quad (14)$$

For explicit full-batch SGD, let $\delta\theta_k = \theta_{k+1} - \theta_k$ and define exact segment kernels

$$\tilde{J}_k = \int_0^1 J(\theta_k + s\delta\theta_k) ds, \quad \tilde{\Theta}_k = \tilde{J}_k J(\theta_k)^\top, \quad (15)$$

$$\tilde{m}_k = \int_0^1 \nabla M(\theta_k + s\delta\theta_k) ds, \quad \tilde{c}_k = J(\theta_k) \tilde{m}_k. \quad (16)$$

The true increments are exactly $r_{k+1} - r_k = \eta \tilde{\Theta}_k q(r_k)/n$ and $M_{k+1} - M_k = \eta \tilde{c}_k^\top q(r_k)/n$. If $\eta > 0$ and, for every $0 \leq j < k$, $\|\tilde{\Theta}_j - \Theta_0\| \leq D_\Theta$, $\|\tilde{c}_j - c_0\| \leq D_c$, and $\|q(r_j)\|, \|q(\bar{r}_j)\| \leq V$, then the explicit-Euler version of Eq. (11) obeys

$$\|r_k - \bar{r}_k\| \leq \frac{D_\Theta V}{n} \Phi_k, \quad (17)$$

$$|M_k - \bar{M}_k| \leq \frac{k\eta D_c V}{n} + \frac{\eta \|c_0\| D_\Theta V}{4n^2} \Psi_k, \quad (18)$$

where

$$\Phi_k = \frac{(1 + \eta a)^k - 1}{a}, \quad \Psi_k = \frac{(1 + \eta a)^k - 1}{\eta a^2} - \frac{k}{a}, \quad (19)$$

with limits $\Phi_k = k\eta$ and $\Psi_k = \eta k(k-1)/2$ at $a = 0$. Apply the bounds separately to B, W and add them to bound the paired response gap. Frozen signs or brackets transfer only when their strict margins exceed the corresponding summed errors; first hitting times additionally require Corollary 10's prehistory, transversality, radius, and mesh hypotheses.

Theorem 5 is conditional: this paper does not prove the required instantaneous or secant-kernel movement bounds for tanh or GRU. A held-out event-classification result cannot substitute for that premise. In particular, the theorem is finite-width and is not a DMFT closure.

For channel-localized modes of the dense linear RNN, differentiation across the disjoint (c, B, W) blocks gives the exact finite-width geometry

$$G_{ab} = p_a^\top p_b + \mathbf{1}\{\text{ch}(a) = \text{ch}(b)\} q_a^\top q_b + \langle S_a, S_b \rangle_F, \quad (20)$$

where $p_a = W^{\ell_a} b_a$, $q_a = (W^{\ell_a})^\top c$, and

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (21)$$

The proofs are in Appendix A. Equation (7) is exact for the zero-background-noise linear synthetic model. For tanh/GRU probes, we use Theorem 1's direct autograd drift as primary and report R ; projected and matched-state decompositions are diagnostics only.

5 From rate suppression to outcome starvation

Theorem 6 (Necessary and sufficient tail-area criterion). *Let Δ be absolutely continuous on $[0, H]$, with $\Delta(0) = 0$. Suppose there is $\tau_d \in (0, H)$ such that $d > 0$ almost everywhere on $(0, \tau_d)$ and $d < 0$ almost everywhere on (τ_d, H) . Define*

$$A_+ = \int_0^{\tau_d} d(u) du = \Delta(\tau_d) > 0, \quad N(t) = \int_{\tau_d}^t -d(u) du. \quad (22)$$

Then $\Delta(t) = A_+ - N(t)$ for $t \geq \tau_d$. Hence equality is reached by H if and only if $N(H) \geq A_+$, and strict outcome starvation occurs by H if and only if $N(H) > A_+$. If $d < 0$ on each nontrivial post-crossing interval, the later equality is unique whenever it exists.

This result blocks a common but false inference: a $+$ $\rightarrow$ $-$ drift crossing locates a local maximum of Δ ; it does not force Δ to return to zero. The post-crossing negative area must repay the positive area already accumulated. Causal starvation further requires weak-only learnability.

Corollary 7 (Response-crossing bounds). *Under Theorem 6, if $d(t) \leq -\kappa < 0$ on $[t_0, H]$ with $t_0 > \tau_d$, then, when the right side lies within the horizon,*

$$\tau_{\text{response}} \leq t_0 + \frac{[A_+ - N(t_0)]_+}{\kappa} \leq t_0 + \frac{A_+}{\kappa}. \quad (23)$$

If $d(\tau_d) = 0$ and $d'(t) \leq -\lambda < 0$ on $[\tau_d, \tau_d + r]$, then $N(t) \geq \lambda(t - \tau_d)^2/2$; when $A_+ \leq \lambda r^2/2$,

$$\tau_{\text{response}} \leq \tau_d + \sqrt{2A_+/\lambda}. \quad (24)$$

5.1 An ordering-invariant rank-one criterion

Assume deterministic positive cues $z^B = (\rho, 1)$, $z^W = (0, 1)$, $\rho > 0$, and exact two-mode realization. Define

$$u_B = \rho M_s^B + M_w^B, \quad s_B = \sigma(-u_B), \quad (25)$$

$$u_W = M_w^W, \quad s_W = \sigma(-u_W), \quad (26)$$

$$H_B = \rho G_{ws}^B + G_{ww}^B, \quad H_W = G_{ww}^W, \quad (27)$$

$$Q_B = \rho^2 G_{ss}^B + 2\rho G_{sw}^B + G_{ww}^B \geq 0. \quad (28)$$

Theorem 8 (Log drift-ratio identity). *The weak drifts factor as $F_w^B = s_B H_B$ and $F_w^W = s_W H_W$. Where $H_B, H_W > 0$, define*

$$\Psi = \log \frac{F_w^B}{F_w^W} = \log \frac{s_B}{s_W} + \log \frac{H_B}{H_W}. \quad (29)$$

Then $\text{sign}(d) = \text{sign}(\Psi)$ and

$$\Psi' = -s_B(1 - s_B)Q_B + s_W(1 - s_W)H_W + \left(\log \frac{H_B}{H_W} \right)'. \quad (30)$$

Corollary 9 (Unique rate crossing). *Suppose on $[t_0, H]$ that $H_B, H_W > 0$, $\Psi(t_0) > 0$, $\Psi'(t) \leq -\kappa < 0$, and $H - t_0 \geq \Psi(t_0)/\kappa$. Then exactly one rate crossing occurs in $(t_0, H]$ and*

$$\tau_d \leq t_0 + \Psi(t_0)/\kappa. \quad (31)$$

A sufficient primitive condition is

$$s_B(1 - s_B)Q_B \geq a, \quad s_W(1 - s_W)H_W \leq b, \quad (\log(H_B/H_W))' \leq c, \quad a - b - c \geq \kappa > 0. \quad (32)$$

The bounds must be established independently to be predictive. Evaluating Ψ on a completed trajectory is certification, not prediction. Noisy cues generally break the rank-one factorization, and even a rate crossing still requires Theorem 6 for an outcome crossing.

5.2 Hitting-time stability

Corollary 10 (Transverse first-hit stability). *Let continuous $x_N, x : [0, H] \rightarrow \mathbb{R}$ satisfy $\|x_N - x\|_\infty \leq \epsilon$, and define*

$$\tau = \inf\{t \in [0, H] : x(t) \geq \beta\}, \quad \tau_N = \inf\{t \in [0, H] : x_N(t) \geq \beta\}, \quad \inf \emptyset = \infty. \quad (33)$$

Assume x is absolutely continuous, $x(0) < \beta$, and $\tau \in (0, H)$. Choose $0 < r \leq \min\{\tau, H - \tau\}$ such that $x'(t) \geq \kappa > 0$ almost everywhere on $[\tau - r, \tau + r]$, and let

$$\eta = \beta - \max_{0 \leq t \leq \tau - r} x(t) > 0. \quad (34)$$

If $\epsilon < \eta$ and $\epsilon/\kappa \leq r$, then $\tau_N < \infty$ and

$$|\tau_N - \tau| \leq \epsilon/\kappa. \quad (35)$$

For an ordered grid $0 = t_0 < \dots < t_m = H$ of maximum mesh $h = \max_j(t_j - t_{j-1})$, let τ_N^{samp} be the first sampled time with $x_N(t_j) \geq \beta$, and let τ_N^{int} be the first hit of the piecewise-linear interpolant through the samples. If $\epsilon/\kappa + h \leq r$, both are finite and

$$|\tau_N^{\text{samp}} - \tau| \leq \epsilon/\kappa + h, \quad |\tau_N^{\text{int}} - \tau| \leq \epsilon/\kappa + h. \quad (36)$$

For a paired delay, if both reference hits are finite and each trajectory satisfies its own assumptions, then

$$|(\tau_{N,B} - \tau_{N,W}) - (\tau_B - \tau_W)| \leq \epsilon_B/\kappa_B + \epsilon_W/\kappa_W + h_B + h_W. \quad (37)$$

If either hit is infinite, the delay is censored; a common mesh h gives the special case with $2h$.

Descending crossings follow by sign reversal and localization after a strict excursion. Since the response gap has $\Delta(0) = 0$, its later downward outcome crossover must be localized to $[a, H]$ with $\Delta(a) > 0$ and treated using $x = -\Delta$ and target zero; the hit at time zero is not that later crossover. This corollary transfers an independently proved uniform trajectory limit. It does not prove that limit, first-entry semantics, the prehistory margin, or the derivative bound. Thus Theorem 5's summed paired response bound licenses a crossing-time statement only after all of these separate guards hold; an error formula or empirical event match alone is not a hitting-time certificate.

6 A solvable zero-disorder wide limit

Set recurrent gain $g = 0$, lag separation zero, and background noise zero in the dense linear model. Write the input columns as b_s, b_w . The recurrent block is inert, $M_a = c \cdot b_a$. Let $u_{ab} = b_a \cdot b_b$ and $w = c \cdot c$.

Theorem 11 (Every-width six-scalar closure). *At every width,*

$$\dot{M}_a = g_a w + \sum_b g_b u_{ab}, \quad \dot{u}_{ab} = g_a M_b + g_b M_a, \quad (38)$$

$$\dot{w} = 2 \sum_b g_b M_b, \quad G_{ab} = u_{ab} + \delta_{ab} w. \quad (39)$$

Theorem 12 (Finite-horizon deterministic wide limit). *Let all coordinates of b_s, b_w, c be mutually independent $\mathcal{N}(0, 1/N)$ and*

$$q_N = (M_s, M_w, u_{ss}, u_{sw}, u_{ww}, w), \quad q_0 = (0, 0, 1, 0, 1, 1). \quad (40)$$

For every $\epsilon > 0$,

$$\mathbb{P}(\|q_N(0) - q_0\|_\infty > \epsilon) \leq \min \left\{ 1, \frac{9}{N\epsilon^2} \right\}. \quad (41)$$

Fix a cue law with bounded second moments and define

$$g_a(M) = \mathbb{E}[z_a \sigma(-z \cdot M)], \quad A = \max_a \mathbb{E}|z_a|, \quad C = \frac{1}{4} \max_a \sum_b \mathbb{E}|z_a z_b|. \quad (42)$$

Then $\|g(M)\|_\infty \leq A$ and

$$\|g(M) - g(\widetilde{M})\|_\infty \leq C \|M - \widetilde{M}\|_\infty. \quad (43)$$

Let $q(t)$ be the deterministic solution from q_0 , let $c_\star = W(e^{-1}) = \sup_x x\sigma(-x)$, where W is Lambert's function, and set

$$R_H = 3 + 4c_\star H, \quad B_{H,r} = R_H + r, \quad L_{H,r} = 4(A + B_{H,r}C) \quad (44)$$

for a fixed deterministic radius $r > 0$. The reference Gram energy $R = w + u_{ss} + u_{ww}$ obeys $R(0) = 3$, $R' \leq 4c_\star$, and therefore $\|q(t)\|_\infty \leq R_H$ on $[0, H]$. The vector field is $L_{H,r}$ -Lipschitz in infinity norm on the radius- r tube about this path. If

$$\|q_N(0) - q_0\|_\infty \leq e^{-L_{H,r}H} \min\{r, \delta\}, \quad (45)$$

then q_N does not leave the tube before H and

$$\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty \leq \delta. \quad (46)$$

Consequently,

$$\mathbb{P} \left(\sup_{t \leq H} \|q_N(t) - q(t)\|_\infty > \delta \right) \leq \min \left\{ 1, \frac{9e^{2L_{H,r}H}}{N \min\{r, \delta\}^2} \right\}. \quad (47)$$

For $r = 1$ and $0 < \delta \leq 1$, the denominator reduces to $N\delta^2$ with the deterministic constant $L_{H,1}$. Thus $q_N \rightarrow q$ uniformly on fixed horizons in probability, with initialization scale $O_{\mathbb{P}}(N^{-1/2})$. For the paired both/weak-only result, use a common r and the maximum of the two deterministic $L_{H,r}$ constants for the two fixed cue laws.

This theorem is intentionally singular: it is a genuine large-width result only at zero disorder and zero lag. It is not a positive-disorder DMFT.

7 Counterfactual Drift Correction

Flatten the implemented trainable tensors into one fixed parameter chart with the standard product Euclidean/Frobenius inner product. Every gradient, dot product, norm, projection, Lipschitz ball, and straight segment in this section uses that chart and metric; no reparameterization-invariance or natural-gradient claim is made.

At a both-feature state θ , let

$$a = \nabla m_s(\theta), \quad p = \nabla m_w(\theta), \quad v = -\nabla L(\theta), \quad F_s = a \cdot v, \quad F_w = p \cdot v. \quad (48)$$

A same-initialization weak-only shadow supplies F_w^W . Define

$$q = \begin{cases} p - \frac{p \cdot a}{\|a\|^2} a, & a \neq 0, \\ p, & a = 0, \end{cases} \quad \delta = [F_w^W - F_w]_+, \quad \alpha = \begin{cases} \delta / \|q\|^2, & \delta > 0, q \neq 0, \\ 0, & \delta = 0. \end{cases} \quad (49)$$

The corrected velocity is $v' = v + \alpha q$. A positive deficit with $q = 0$ is infeasible. The theorems assume full-batch gradients, plain SGD, zero weight decay, and no clipping. A nonbinding coefficient cap leaves the uncapped result unchanged; a binding cap preserves orthogonality but generally loses target attainment, so no target-attaining optimality claim then applies.

Theorem 13 (CDC local guarantees). *The construction in Eq. (49) has the following fixed-coordinate properties.*

1. **Instantaneous preservation:** $a \cdot v' = a \cdot v$ exactly, including when α is capped.
2. **Minimum Euclidean norm:** for the feasible uncapped problem, $d^* = \alpha q$ is the unique minimum-norm, lower-bound-attaining correction satisfying $a \cdot d = 0$ and $p \cdot (v + d) \geq F_w^W$.
3. **Local finite-step bound:** if ∇m_s is L -Lipschitz in the same norm on a convex neighborhood containing both straight coordinate step segments, then

$$|m_s(\theta + \eta v') - m_s(\theta + \eta v)| \leq \frac{L}{2} \eta^2 (\|v'\|^2 + \|v\|^2). \quad (50)$$

For the active uncapped correction, if $\|v\| \leq V$, $\delta \leq D$, and $\|q\| \geq q_{\min} > 0$, the right side is at most

$$\frac{L}{2} \eta^2 [V^2 + (V + D/q_{\min})^2]. \quad (51)$$

This explicit norm bound also remains valid under a nonnegative upper cap, which can only reduce α .

For a positive raw deficit, exact feasibility can be stated intrinsically only as dm_w having a nonzero restriction to $\ker(dm_s)$; in this metric it is equivalent to $q \neq 0$. The represented q , its norm, the numerical `feasibility_epsilon`, and the selected minimum-norm correction are metric- and parameter-scale-dependent. The finite-step result is one-step and local in the same chart. It is not a trajectory guarantee and does not imply final strong-response retention.

8 Executable theorem contracts

The implementation evaluates theorem formulas and exposes scope metadata. It enforces the scalar guards represented in helper arguments or trainer configuration; callers still establish trajectory, model, first-entry, and smoothness hypotheses. Table 1 lists the primary contracts. Unit tests check these software identities and edge cases; tests are not mathematical proofs.

Every general positive-disorder/positive-lag solver path that would require an unproved closure raises `NotImplementedError`. This is a scientific guard, not missing numerical polish.

Table 1: Executable contracts in `src/gradient_starvation/theory.py`.

Contract	Mathematical object	Scope guard
<code>projected_statistics</code>	direct Eq. (6), Gg , residual	direct drift primary for nonlinear probes
<code>initial_frozen_empirical_kernel</code>	J_0, Θ_0, c_0 in Theorem 3	initialization only; full sample kernel
<code>integrate_frozen_logistic_sgd</code>	explicit Euler for Eq. (11)	exact for tangent surrogate, not nonlinear model
<code>frozen_kernel_discrete_error_bound</code>	Eq. (18)	caller supplies secant-kernel movement bounds
<code>equal_time_drift_difference</code>	Δ' and two projected product orderings	exposes non-canonical attribution
<code>rank_one_drift_ratio</code>	Theorem 8 factors and Ψ	caller supplies noiseless/exact hypotheses
<code>discrete_crossover_certificate</code>	telescoping finite-step tail areas	learnability gate required for causal label
<code>transverse_hitting_time_error_bound</code>	$\epsilon/\kappa + h$	guards supplied error, pre-hit separation, radius, mesh; caller proves first entry and derivative
<code>zero_disorder_failure_bound</code>	Eqs. (41), (47)	caller supplies deterministic reference-tube $L_{H,r}$ and bootstrap radius r
<code>cdc_finite_step_deviation_bound</code>	Eq. (50)	fixed-coordinate Euclidean evaluator; caller supplies local smoothness and segment containment

Table 2: Final synthetic experiment configurations. “Shadow” methods train a second weak-only model and therefore use roughly twice the model memory and per-step compute.

Experiment	Model / width	Task grid	Samples	Steps / LR	Seeds	Primary purpose
E1	dense linear / 96	$\rho \in \{1, 2, 4, 8\}$, lag $\in \{0, 2, 4, 8\}$	256	500 / .01	4–11	causal phase grid
E2	dense linear / 32–256	$(1, 0), (2, 2), (4, 4), (8, 8)$	1024	400 / .01	0–7	held-out closure convergence
E-NL	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	20–27	direct-drift trajectory certificate
NTK pilot	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	1×8	five-output falsification
NTK factorial	tanh, GRU / 32	$\rho = 4$, lag 2	256	1000 / .01	4×8	crossing-event validation
E3	dense linear / 96	$\rho = 4$, lag 2	256	500 / .01	40–47	eight-method CDC ablation

9 Experimental protocol

9.1 Synthetic task and pairing

Sequences have two input channels. A strong cue of scale ρ and a unit weak cue appear at controlled sequence positions separated by a lag. Positive tasks align the cues with the label; the negative control reverses the strong cue. Data and initialization are paired across both-feature and weak-only conditions. Training is deterministic full-batch cross-entropy with zero weight decay. Except for E-NL, the two conditions can be run sequentially because deterministic full-batch updates preserve the same trajectories; E-NL uses lockstep models so both exact drifts are available at each logged time.

E1 uses gain 0.5, $\beta = 0.25$, horizon $H = 5$, and a single eight-seed negative-control cell $(\rho, \text{lag}) = (4, 4)$. E2 uses gain 0.5 and a held-out empirical closure reference with width 384, seeds 100–115, 16 particles, and 32,768 field samples. E-NL uses gain 0.8, cue noise 0.1, zero background noise, $\beta = 0.5$, and direct autograd drift at every logged point. The frozen- kernel studies use the same task cell. An earlier engineering set (data seed 100, model seeds 1000–1007) is excluded: a first evaluator grid mismatch stopped before training, and a second grid mismatch stopped only after nonlinear outcomes had been generated. No scores were inspected. The set was used solely to repair CSV-grid alignment (integer steps plus a numerical tau check); it did not change labels, acceptance thresholds, or the five pilot endpoints. The valid broad pilot therefore uses a fresh unseen data seed and eight unseen model seeds; after it fails, the restricted validation uses fresh data seeds 201–204 crossed with model seeds 2010–2017. E3 compares ERM, Spectral Decoupling, an interaction penalty, CDC, loss-gradient projection, unconstrained rescue, Bloop-style, and PCGrad-style methods. The five rescue methods share the same weak-only shadow target and differ in what they protect.

Manifest-sealed prediction protocol. `enl-preflight` constructs no optimizer and observes no trained state. It writes the initialization kernels and predictions, verifies state/gradient nonmutation, and hashes every declared file and per-record kernel together with the resolved config and executable-source fingerprint. Before constructing an optimizer, `enl-evaluate` requires the sidecar, actual manifest, and externally pinned digest to agree; verifies every hash/size, source, protocol, seed pair, and time grid; and loads rather than rebuilds predictions. It re-hashes all sealed inputs after training. The broad five-endpoint rule was frozen before its pilot; after that rule failed, only drift/response crossing was frozen as primary on fresh seeds. The latter crossed design uses a two-way pigeonhole bootstrap with 10,000 replicates.

Statistics and censoring. Independent seed pairs are the unit of replication only when neither seed axis is reused. Under crossed data/model-seed reuse, row-wise Student- t intervals are suppressed and the

reported two-way percentile interval resamples the observed two seed axes independently. It is conditional on those observed seed levels; in particular, the all-correct response table necessarily gives $[1, 1]$ and does not quantify unseen failure modes. Hitting times beyond the horizon are right-censored and never converted into finite delays. Conditional crossing-time means are reported only with their number of crossing records. The exploratory E-NL 5×5 grid uses seeds 0–3 and is explicitly post-hoc; it is not used to select a confirmatory condition.

Reproducibility. Ordinary trajectory runs write a resolved YAML, environment record, trajectories, summary, aggregate tables, and figures. Frozen preflights instead write predictions, initialization kernels, an acceptance record, and a manifest; evaluations write observed trajectories, per-record scores, metrics, acceptance, and provenance. The final E-NL rerun is reconstructible from its later frozen repository source at commit `2d0ee83` / tag `theorem-aligned-v1`. The two frozen studies record source fingerprints, but their exact dirty source bytes were not archived; the current tree therefore fails their deliberate source guard and cannot replay them bit-for-bit. Their compact archives support an audit of the reported outcomes, seals, and configurations, not executable regeneration. Historical E1, E2, E2-R, exploratory E-NL, and E3 artifacts likewise record dirty Git states without reconstructible executed source.

10 Results

10.1 Controlled dense-linear phase grid

At lags 0, 2, and 4, all 32 condition–seed pairs per lag have positive weak AUC gaps. Averaged across ρ , the lag-wise gaps are 5.76, 8.93, and 6.11, respectively; the effect peaks at lag 2 rather than decreasing monotonically with lag. At every fixed lag, the gap increases monotonically over the measured ρ grid. The negative-control cell is transfer in 8/8 seeds, with mean gap -4.038 and 95% CI $[-4.498, -3.578]$. Every lag-8 cell is weak-only unlearnable, so these 32 rows are not labeled starvation. Final accuracy is 1.0 throughout, and no row meets the target at initialization.

10.2 Finite-width agreement and exact special cases

E2 compares finite networks with a disjoint-seed empirical *closure reference*; it is not an independently derived theory. Overall both-condition trajectory NRMSE is 0.09084, 0.04761, 0.05149, 0.02958 at widths 32, 64, 128, 256. Weak-only NRMSE is 1.56931, 0.38792, 0.40201, 0.20024. Both improve end-to-end but are non-monotone at $N = 128$. All 32 point–condition–diagnostic combinations improve end-to-end; large normalized weak-only values at difficult points have near-zero denominators. The exact finite-width geometry check has zero recorded relative error and the maximum projected-identity error is 2.86×10^{-6} .

In the exact zero-disorder solver check, the time-discretization residual scales as $O(\eta)$ with fitted slope 1.0115. A frozen-geometry integrator agrees with the projected-flow integrator to 1.91×10^{-15} . The general solver acceptance record remains false by construction because all general-DMFT checks are blocked.

10.3 Nonlinear crossover is seed-dependent

Table 3 is the authoritative nonlinear result. Tanh has a complete exact-drift/response/tail-area causal certificate in 3/8 seeds; the other five have negative exact relative drift from initialization and therefore do not exhibit transfer-to-starvation. GRU has exact drift and response crossings in 5/8 seeds, but the weak-only trajectory fails to reach $\beta = 0.5$ in all eight seeds; its causal count is therefore zero. The nonzero direct-versus-projected residuals empirically confirm that Gg cannot be silently treated as the exact nonlinear probe derivative.

The post-hoc tanh search over $\rho \in \{1.5, 2, 3, 4, 6\}$ and lags $\{0, 1, 2, 4, 8\}$ found no cell with 4/4 causal certificates; the best cells reached 3/4. We retain this as a negative search audit and do not select a searched cell for a paper claim. In particular, the superseded 8/8 tanh result and its $\tau^* = 1.6112$ used a different response convention and are not valid under the final definition.

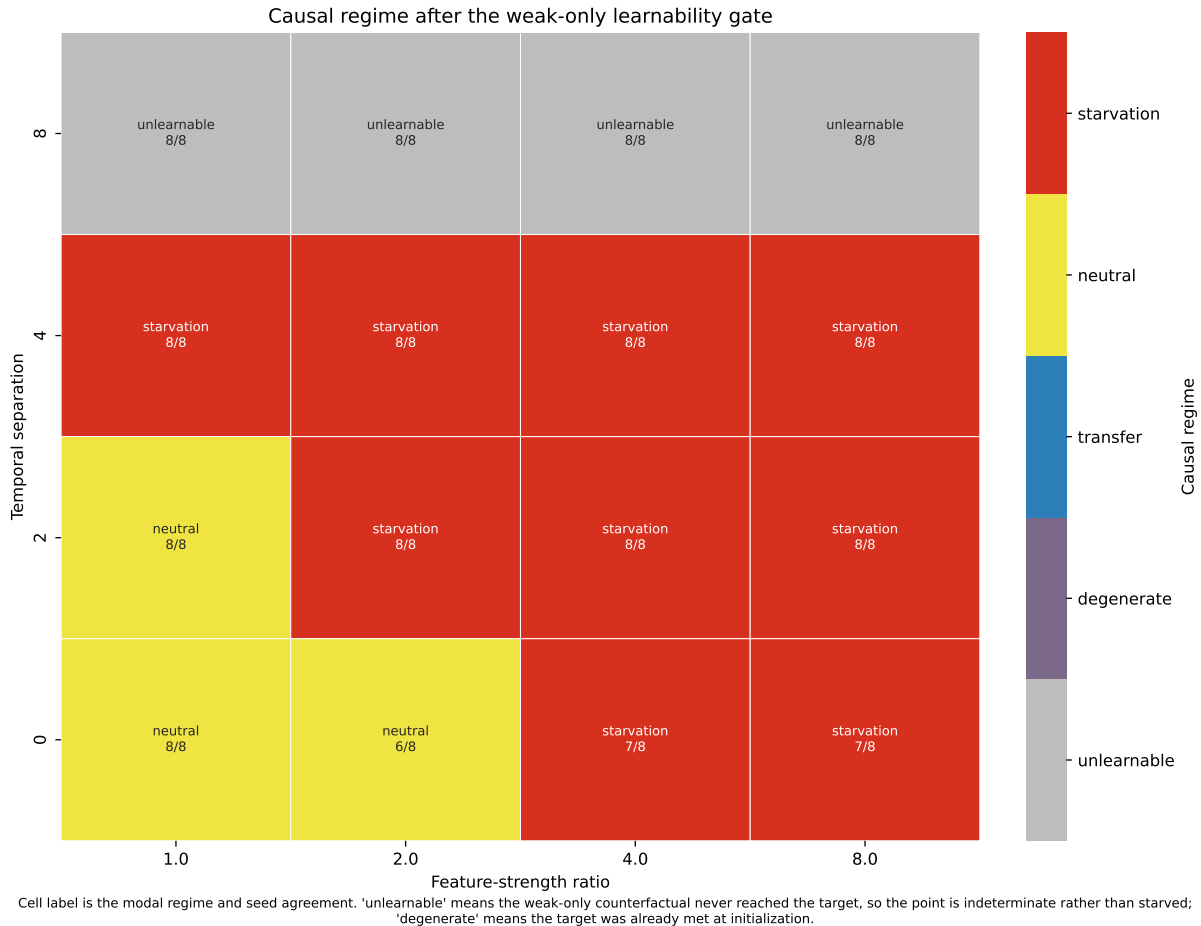


Figure 1: Controlled dense-linear causal regions. The learnability gate prevents lag-8 points from being mislabeled as starvation.

10.4 Frozen kernels predict crossing events, not broad starvation phase

The initialization-frozen predictor was first tested under a sealed five-output pilot using one unseen data seed and eight unseen model seeds per architecture. It obtained 12/16 phase, 14/16 drift-crossing, 14/16 response-crossing, 14/16 causal- certificate, and 12/16 weak-learnability classifications. The preregistered rule nevertheless *failed*: tanh learnability was 4/8, below the 5/8 per-model floor. We preserve that rejection rather than averaging it away.

Only after this failure did we define the narrower crossing-event hypothesis. A fresh sealed factorial crossed four unseen data seeds with eight unseen model seeds for each architecture. Table 4 reports classification correctness, not observed crossing prevalence. The restricted rule passed: all response-crossing labels and 60/64 drift-crossing labels were correct; the two-way bootstrap lower endpoints exceeded the frozen 0.5 floor.

The restriction is substantive. In the factorial, phase, causal-certificate, and learnability correctness were only 52/64, 56/64, and 49/64 and were explicitly secondary; tanh learnability was 17/32. Crossing times remained systematically early (overall drift/response bias $-0.822/-1.189$), and tanh response-gap and weak- response RMSE were 2.455 and 2.489 despite high sign agreement. Thus the evidence supports an uncalibrated event classifier at this one width/task cell. It neither proves Theorem 5's kernel premise nor validates phase, causal starvation, learnability, trajectory magnitude, calibrated time, or prevalence.

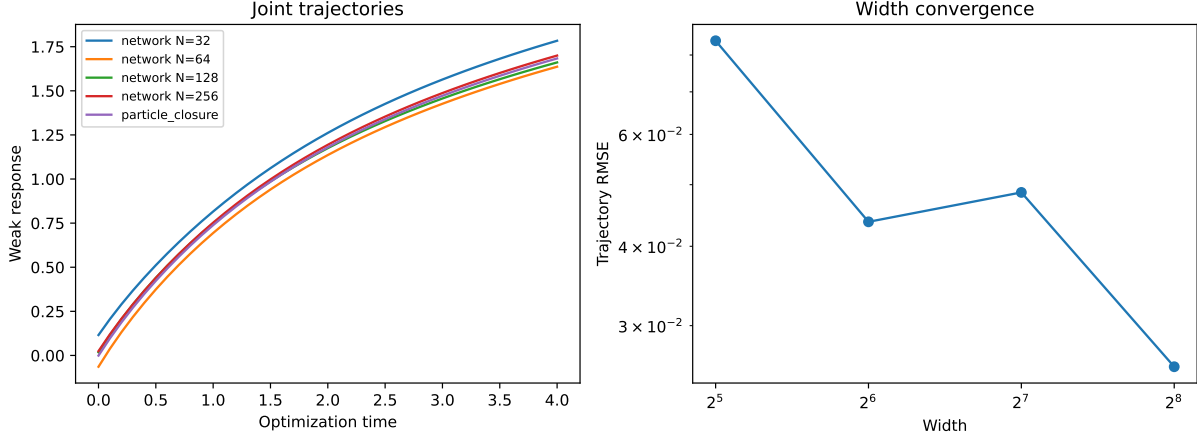


Figure 2: Finite-width agreement with a held-out empirical finite-network closure reference. This is not DMFT validation.

Table 3: Final E-NL result at $\rho = 4$, lag 2, eight seeds. Crossing times and confidence intervals are conditional on crossing and must not be read as eight-seed estimates.

Model	exact drift	response	causal	mean τ_d	mean τ_r	max $ R_w^B - R_w^W $
tanh	3/8	3/8	3/8	0.194	0.389	0.02197
GRU	5/8	5/8	0/8	3.846	7.200	0.002986

10.5 CDC has the local guarantee, not the best final tradeoff

All five shadow-based methods reduce the causal weak AUC gap from ERM’s 10.5155 to approximately zero and attain their shared weak-drift target at every logged step. Their weak outcomes are numerically similar to roughly four decimal places, but not statistically equivalent: 9/10 direct paired comparisons are significant at 0.05, while the differences are practically tiny relative to the roughly 10.5 change from ERM. CDC alone preserves instantaneous strong drift to rounding scale, 4.77×10^{-7} , versus 1.48×10^{-1} to 3.75×10^{-1} for the alternative shadow constraints. This local property does not yield the best final strong response: unconstrained rescue and the PCGrad-style arm end at 1.295, compared with CDC’s 1.042 and ERM’s 1.364.

11 Discussion and limitations

What is proved. The finite-width cross-kernel identity, exact full sample-logit/response dynamics, initialization-frozen tangent surrogate, conditional continuous/secant-SGD comparison, dense-linear geometry, tail-area equivalence, conditional rank-one crossover criterion, quantitatively isolated transverse hitting-time bound, singular zero-disorder wide limit, and fixed-coordinate local CDC results are mathematical statements with proofs. The frozen comparison does not prove its kernel-movement premises. Executable contracts reproduce the algebra and enforce represented scalar guards; callers establish remaining trajectory and model hypotheses.

What the experiments establish. The controlled dense model shows robust positive AUC gaps in the learnable region. The realized nonlinear trajectories do not establish a universal crossover: tanh is seed-dependent, and GRU fails causal learnability despite response crossings. The broad frozen-kernel pilot also failed. On fresh seeds, its post-pilot restriction predicts drift/response crossing events well at one controlled cell, but not broad phase or causal starvation; timing is early and tanh magnitude is poor. The exploratory search did not find a 4/4 robust cell. CDC demonstrates its exact local constraint but not a superior practical tradeoff.

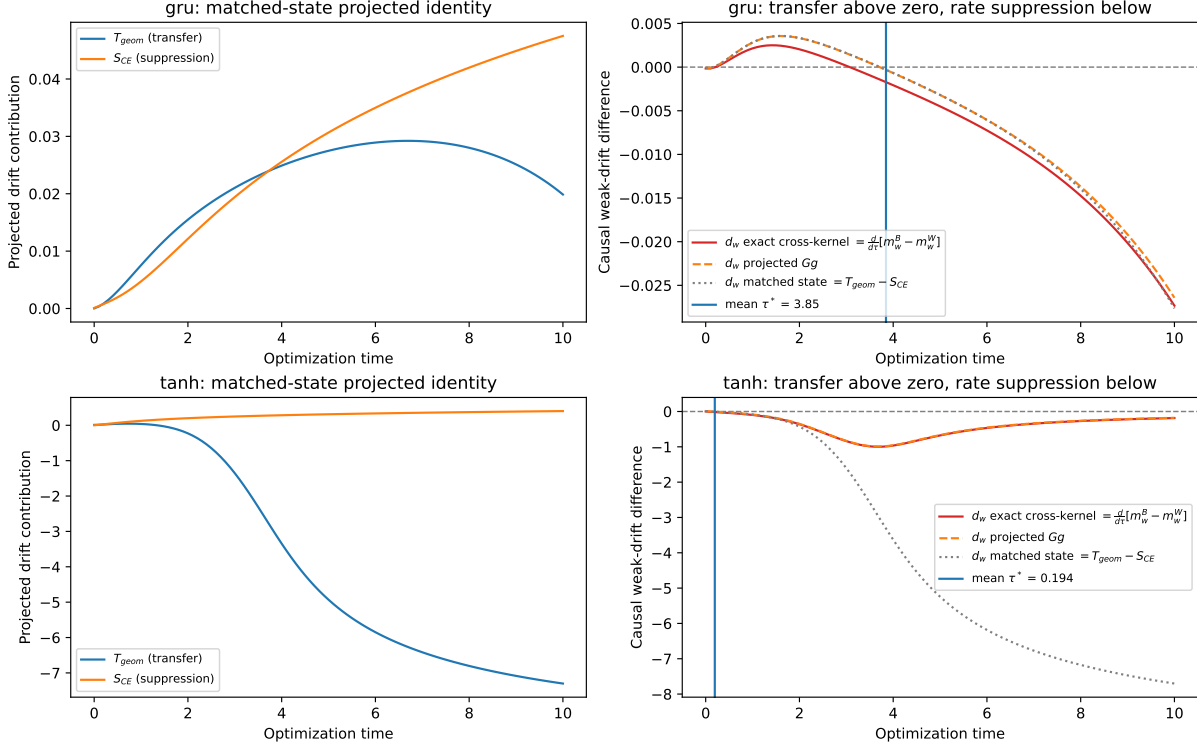


Figure 3: Theorem-aligned nonlinear trajectories. Direct-autograd drift is primary; projected and matched-state quantities are separately styled diagnostics.

Table 4: Manifest-sealed frozen-kernel evaluations. The broad pilot’s complete five-output rule failed. Only the two event rows in the fresh factorial were preregistered primary endpoints. CIs resample data- and model-seed axes.

Protocol	Endpoint	Overall	tanh	GRU	95% CI
Broad pilot	phase	12/16	6/8	6/8	diagnostic
Broad pilot	weak learnability	12/16	4/8	8/8	failed
Restricted 4×8	drift event	60/64	32/32	28/32	[0.8125, 1]
Restricted 4×8	response event	64/64	32/32	32/32	[1, 1]

Open mean-field theory. For positive recurrent disorder or positive lag, a joint optimization-time DMFT would need (i) an effective single-site process and covariance/response kernels, (ii) all visible and loss-invisible closure variables, (iii) finite-horizon existence and uniqueness, (iv) joint quenched concentration under shared disorder, and (v) discretization and hitting-time control. Neither fitting an empirical closure reference nor passing an event-classification test supplies these objects or proves kernel stability. We therefore leave an analytic ρ_c and certified or quantitatively accurate tanh/GRU crossing-time prediction open.

Data and external validity. All citable results are controlled synthetic experiments. A Waterbirds adapter contains an unrun CDC-style minibatch head-coordinate surrogate, but it lies outside the full-batch matched-shadow theorem. Its proposed weak coordinate is an intercept rather than an identified bird-shape response, and no exact background ablation exists; we make no Waterbirds claim. If pursued, the preregistered sensitivity pilot must precede a full run and oracle/group-agnostic information settings must be separated.

Reproducibility and baselines. The final E-NL rerun is content-addressed and compactly archived at repository commit 2d0ee83 / tag theorem-aligned-v1. Its recorded digest identifies the executed run

Table 5: Dense CDC ablation, eight paired seeds. Shadow-based arms have access to a matched weak-only model; the other arms do not. “Strong change” is the maximum absolute instantaneous strong-drift perturbation.

Method	Shadow	Weak AUC gap	Strong change	Final m_s
ERM	no	10.5155	—	1.364
Spectral Decoupling	no	9.0229	—	0.829
Interaction	no	10.5145	—	1.365
CDC	yes	−0.0005	4.77×10^{-7}	1.042
Loss-gradient projection	yes	−0.0002	3.75×10^{-1}	0.371
Unconstrained rescue	yes	−0.0007	1.48×10^{-1}	1.295
Bloop-style	yes	0.0002	3.75×10^{-1}	0.370
PCGrad-style	yes	−0.0007	1.48×10^{-1}	1.295

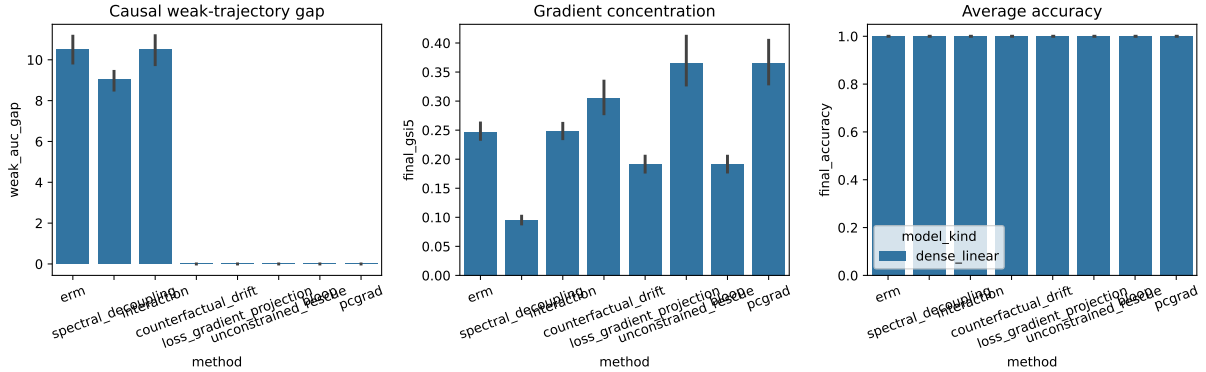


Figure 4: Eight-method dense-linear mitigation ablation. CDC’s established distinction is instantaneous first-order preservation, not superior weak rescue or final strong retention.

source, not later descendants or working trees. Historical E1, E2, E2-R, exploratory E-NL, and E3 runs retain a disclosed dirty-source provenance limitation. The Bloop-style and PCGrad-style baselines are repository implementations motivated by the cited methods, not yet independently audited replicas. CDC also has a stronger supervision and compute setting than ERM, Spectral Decoupling, and the interaction penalty.

Submission readiness. The package is scientifically defensible as a controlled theory paper, but the seed-dependent nonlinear headline, lack of real-data evidence, historical dirty-run provenance, and unaudited external baselines limit its present empirical strength. No theorem or experiment can guarantee an oral acceptance outcome.

12 Conclusion

Gradient starvation requires more than a decreasing weak-feature rate. In a paired causal comparison, a rate crossover becomes an outcome crossing exactly when its negative tail area repays the earlier transfer advantage, and a causal label also requires weak-only learnability. This distinction turns an ambiguous mechanism into a theorem and an executable certificate. The same discipline narrows the rest of the paper: $\dot{M} = Gg$ is exact only under explicit realization assumptions; the rank-one criterion predicts only when its bounds are independently established; the wide limit is singular; the frozen empirical-NTK comparison is conditional on unproved kernel movement; and CDC’s exact guarantee is local. The corrected nonlinear results

are correspondingly narrow: the broad predictor failed, while a fresh factorial supports only drift/response-crossing event classification at one controlled cell. Establishing a general recurrent optimization-time theory, causal predictor, or calibrated crossing time remains open rather than a numerical placeholder.

A Proofs for the finite-width theorem package

A.1 Proof of Theorem 1 and Corollary 2

For $\ell(r) = \log(1 + e^{-r})$, $\ell'(r) = -\sigma(-r)$. Therefore

$$\dot{\theta}_C = -\nabla L_C = \frac{1}{n_C} \sum_i \sigma(-r_i^C) \nabla r_i^C. \quad (52)$$

The chain rule gives

$$\dot{M}_a^C = \langle \nabla M_a^C, \dot{\theta}_C \rangle = \frac{1}{n_C} \sum_i \langle \nabla M_a^C, \nabla r_i^C \rangle \sigma(-r_i^C), \quad (53)$$

which proves Theorem 1. Under exact realization, $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C$. Substitution and exchange of finite sums yield Eq. (7). If an orthogonal residual logit term is present, substitution of $\nabla r_i^C = \sum_b z_{ib}^C \nabla M_b^C + \nabla r_i^\perp$ instead yields Eq. (8). $\square$

A.2 Proofs of Theorems 3 and 5

The parameter flow above can be written $\dot{\theta} = J^\top q(r)/n$. Therefore the chain rule gives $\dot{r} = J\dot{\theta} = JJ^\top q(r)/n$ and $\dot{M} = \langle \nabla M, \dot{\theta} \rangle = c^\top q(r)/n$, proving Theorem 3.

For the comparison, q is 1/4-Lipschitz componentwise and hence in Euclidean norm. Subtracting Eq. (11) from Eq. (10), adding and subtracting $\Theta_0 q(r)$, and writing $e_r = \|r - \bar{r}\|$ gives

$$e'_r(u) \leq \frac{\|\Theta_0\|}{4n} e_r(u) + \frac{B_\Theta V}{n} = a e_r(u) + \frac{B_\Theta V}{n}. \quad (54)$$

Grönwall with $e_r(0) = 0$ proves Eq. (13). The same add-subtract step gives

$$|\dot{M} - \dot{M}| \leq \frac{B_c V}{n} + \frac{\|c_0\|}{4n} e_r. \quad (55)$$

Integrating and using $\int_0^t \phi_a(u) du = \psi_a(t)$ proves Eq. (14).

For SGD, the fundamental theorem of calculus along the straight update segment gives

$$r_{j+1} - r_j = \tilde{J}_j \delta \theta_j = \frac{\eta}{n} \tilde{\Theta}_j q(r_j), \quad (56)$$

$$M_{j+1} - M_j = \langle \tilde{m}_j, \delta \theta_j \rangle = \frac{\eta}{n} \tilde{c}_j^\top q(r_j). \quad (57)$$

Subtracting the frozen Euler recursion yields scalar recurrences

$$e_{j+1} \leq (1 + \eta a) e_j + \frac{\eta D_\Theta V}{n}, \quad (58)$$

$$E_{j+1} \leq E_j + \frac{\eta D_c V}{n} + \frac{\eta \|c_0\|}{4n} e_j. \quad (59)$$

Solving the first geometric recurrence and summing it in the second gives Eq. (18), including the stated $a = 0$ limits. Paired bounds are the triangle inequality. Error-separated sign transfer follows immediately; first-hit transfer requires the additional assumptions of Corollary 10. $\square$

A.3 Proof of the dense-linear geometry

For a unit response mode in channel a at lag ℓ_a , write b_a for the corresponding input column. Direct differentiation of $c^\top W^{\ell_a} b_a$ gives gradient $p_a = W^{\ell_a} b_a$ with respect to c , gradient $q_a = (W^{\ell_a})^\top c$ with respect to column b_a , and recurrent gradient

$$S_a = \sum_{j=0}^{\ell_a-1} (W^j)^\top c (W^{\ell_a-1-j} b_a)^\top. \quad (60)$$

The parameter blocks are disjoint, so their gradient inner products add. Input columns contribute only when the channels coincide, giving Eq. (20). $\square$

A.4 Proof of Theorem 6 and Corollary 7

Absolute continuity and the fundamental theorem of calculus give

$$\Delta(\tau_d) = \Delta(0) + \int_0^{\tau_d} d = A_+, \quad \Delta(t) = \Delta(\tau_d) + \int_{\tau_d}^t d = A_+ - N(t). \quad (61)$$

The sign assumption makes N continuous and strictly increasing on every nontrivial post-crossing interval. Equality, strict negativity, and uniqueness are therefore respectively equivalent to $N(H) \geq A_+$, $N(H) > A_+$, and strict post-crossing growth of N .

If $d \leq -\kappa$ after t_0 , then $N(t) - N(t_0) \geq \kappa(t - t_0)$; solve for the time needed to cover the remaining area. If $d'(t) \leq -\lambda$ and $d(\tau_d) = 0$, integration gives $d(t) \leq -\lambda(t - \tau_d)$ and then $N(t) \geq \lambda(t - \tau_d)^2/2$. Applying the theorem proves both stated bounds. $\square$

A.5 Proof of Theorem 8 and Corollary 9

The deterministic cue law gives $g^B = s_B(\rho, 1)$ and $g^W = s_W(0, 1)$. Equation (7) therefore gives $F_w^B = s_B H_B$, $F_w^W = s_W H_W$, $u'_B = s_B Q_B$, and $u'_W = s_W H_W$. Since

$$\frac{d}{du} \log \sigma(-u) = -(1 - \sigma(-u)), \quad (62)$$

differentiating Ψ yields Eq. (30). When both drifts are positive, monotonicity of the logarithm gives $\text{sign}(F_w^B - F_w^W) = \text{sign}(\Psi)$.

Under $\Psi' \leq -\kappa$, integration gives $\Psi(t) \leq \Psi(t_0) - \kappa(t - t_0)$. Continuity forces a zero by $t_0 + \Psi(t_0)/\kappa \leq H$, while strict decrease gives uniqueness. The primitive inequalities imply the derivative bound term by term. $\square$

A.6 Proof of Corollary 10

Continuity and first-entry semantics imply $x(\tau) = \beta$. Absolute continuity and $x' \geq \kappa$ almost everywhere give, by integration,

$$x(t) \leq \beta - \kappa(\tau - t) \quad (\tau - r \leq t \leq \tau), \quad x(t) \geq \beta + \kappa(t - \tau) \quad (\tau \leq t \leq \tau + r). \quad (63)$$

On $[0, \tau - r]$, the quantitative prehistory margin and $\epsilon < \eta$ give $x_N < \beta$. On $[\tau - r, \tau - \epsilon/\kappa]$, the left cone and the uniform error again give $x_N < \beta$. Hence $\tau_N \geq \tau - \epsilon/\kappa$. At $t = \tau + \epsilon/\kappa$, the right cone gives $x_N(t) \geq \beta$, so the hit is finite and $\tau_N \leq \tau + \epsilon/\kappa$. This proves Eq. (35).

For sampled data, the first grid point at or after $\tau + \epsilon/\kappa$ is at most h later and, by the radius condition, remains inside the right cone; it is therefore at or above β . No sampled hit can precede the continuous lower bound. If the piecewise-linear interpolant first crosses on $[t_{j-1}, t_j]$, then $x_N(t_j) \geq \beta$, so $t_j \geq \tau - \epsilon/\kappa$ and $t_{j-1} \geq \tau - \epsilon/\kappa - h$; its upper bound is no later than the sampled hit. This proves Eq. (36). Applying the bound separately to two finite hits and using the triangle inequality proves the paired statement. $\square$

A.7 Proof of Theorems 11 and 12

At zero recurrent gain and lag, gradient flow gives $\dot{b}_a = g_a c$ and $\dot{c} = \sum_b g_b b_b$. Differentiating $M_a = c \cdot b_a$, $u_{ab} = b_a \cdot b_b$, and $w = c \cdot c$ proves Eq. (39). The mode gradient has one component in the c block and one in its own b_a block, yielding $G_{ab} = u_{ab} + \delta_{ab} w$.

At initialization, the variances of M_s, M_w, u_{sw} are $1/N$ and those of $u_{ss} - 1, u_{ww} - 1, w - 1$ are $2/N$. Coordinatewise Chebyshev bounds and a union bound sum to $1 + 1 + 1 + 2 + 2 + 2 = 9$, proving Eq. (41).

The field bound $\|g(M)\|_\infty \leq A$ follows from $0 \leq \sigma \leq 1$. Since σ is $1/4$ -Lipschitz,

$$|g_a(M) - g_a(\widetilde{M})| \leq \frac{1}{4} \sum_b \mathbb{E}|z_a z_b| \left\| M - \widetilde{M} \right\|_\infty, \quad (64)$$

which proves Eq. (43). Define the Gram energy $R = w + u_{ss} + u_{ww}$. Along the deterministic flow,

$$R' = 4g \cdot M = 4\mathbb{E}[X\sigma(-X)], \quad X = z \cdot M. \quad (65)$$

Because $x\sigma(-x) \leq c_* = W(e^{-1})$ and $R(0) = 3$, $R(t) \leq R_H$. Every coordinate of q is a squared norm or inner product of b_s, b_w, c , so Cauchy–Schwarz gives $\|q(t)\|_\infty \leq R(t) \leq R_H$.

Consider two states in the cube $\|q\|_\infty, \|\tilde{q}\|_\infty \leq B_{H,r}$ and set $d = \|q - \tilde{q}\|_\infty$. Every product in the M and u equations satisfies the direct estimate

$$|g_a(M)y - g_a(\widetilde{M})\tilde{y}| \leq (A + B_{H,r}C)d. \quad (66)$$

There are at most three such products in an M equation and two in a u equation; the factor two and two cue indices give four in the w equation. Thus the full vector field is $L_{H,r} = 4(A + B_{H,r}C)$ -Lipschitz on the cube and therefore on the radius- r reference tube.

Let T_r be the first time before H that $\|q_N(t) - q(t)\|_\infty$ reaches the tube boundary. Up to T_r , Gronwall's inequality and Eq. (45) give

$$\|q_N(t) - q(t)\|_\infty \leq e^{L_{H,r}t} \|q_N(0) - q_0\|_\infty. \quad (67)$$

If $L_{H,r} > 0$, the initialization event makes this bound strictly less than r for every $t < H$, so the boundary cannot be reached before H . If $L_{H,r} = 0$, then $A = C = 0$, the cue law is zero almost surely, and the six-scalar vector field vanishes; the initial discrepancy is constant and cannot exit the closed tube. The endpoint may lie on the boundary. In either case the same estimate is at most δ throughout $[0, H]$, proving Eq. (46). Applying Eq. (41) with $\epsilon = e^{-L_{H,r}H} \min\{r, \delta\}$ proves Eq. (47). For the shared-initialization pair, use a common r and the maximum deterministic constant from the two fixed cue laws; the same concentrating initial six scalars control both paths. $\square$

B Proofs for Counterfactual Drift Correction

B.1 Instantaneous preservation

If $a = 0$, both strong drifts vanish. Otherwise,

$$a \cdot q = a \cdot p - \frac{p \cdot a}{\|a\|^2} (a \cdot a) = a \cdot p - p \cdot a = 0, \quad (68)$$

so $a \cdot v' = a \cdot v + \alpha a \cdot q = a \cdot v$. This argument is independent of α , hence survives a cap. $\square$

B.2 Unique minimum-norm correction

Let $\bar{\delta} = F_w^W - F_w$ be the raw deficit and $\delta = [\bar{\delta}]_+$. If $\bar{\delta} \leq 0$, $d = 0$ is feasible and uniquely norm-minimal. Suppose $\bar{\delta} > 0$, so $\delta = \bar{\delta}$. Any feasible d obeys $a \cdot d = 0$ and $p \cdot d \geq \delta$. Because $p = q + (p \cdot a / \|a\|^2)a$ when $a \neq 0$ (and $q = p$ when $a = 0$), $p \cdot d = q \cdot d$. Cauchy–Schwarz in the fixed Euclidean subspace orthogonal to a gives

$$\delta \leq q \cdot d \leq \|q\| \|d\|, \quad \|d\| \geq \delta / \|q\|. \quad (69)$$

Equality requires a nonnegative multiple of q . The correction $d^* = (\delta / \|q\|^2)q$ is feasible and attains this lower bound, so it is the unique minimum-norm, lower-bound-attaining feasible correction. Larger positive multiples can also attain the target but have larger norm. A positive deficit is feasible exactly when $q \neq 0$ in this representation. A nonbinding cap leaves d^* unchanged; a binding cap generally loses target attainment, so the target-attaining optimization claim no longer applies. $\square$

B.3 Local finite-step deviation

In the fixed chart and norm, L -Lipschitzness of the gradient on both straight segments gives the smoothness remainder

$$|m_s(\theta + h) - m_s(\theta) - \nabla m_s(\theta) \cdot h| \leq \frac{L}{2} \|h\|^2. \quad (70)$$

Apply this at $h = \eta v$ and $h = \eta v'$. Instantaneous preservation cancels the linear terms and the triangle inequality yields Eq. (50) for the actual velocities, capped or uncapped. For active uncapped CDC,

$$\|v'\| \leq \|v\| + \frac{\delta}{\|q\|} \leq V + D/q_{\min}, \quad (71)$$

which gives the explicit bound; a nonnegative upper cap can only decrease the added multiple of q . If smoothness is known only on the fixed-coordinate ball $B(\theta, r)$, sufficient segment conditions are $\eta V \leq r$ and $\eta(V + D/q_{\min}) \leq r$. These quantities and L are chart- and metric-dependent. $\square$

C Discrete certificate and experiment audit

For logged values Δ_k , increments telescope exactly:

$$\Delta_K - \Delta_0 = \sum_{k=0}^{K-1} (\Delta_{k+1} - \Delta_k). \quad (72)$$

The executable certificate requires one strict block of positive increments followed by one strict block of negative increments. The peak-minus-initial value is its positive area; peak-minus-final is its negative tail area. It certifies causal starvation only if the final trajectory is strictly below the initial gap and the external weak-only learnability flag is true. This finite-step certificate makes no assertion between logged points.

Table 6: Authoritative artifact map. Paths are relative to the repository root.

Claim family	Artifact directory
Controlled dense phase grid	results/e1_dense_rerun-20260823-085142
Finite-width empirical closure	results/e2_corrected_validation-20260821-224630
Exact special solver checks	results/e2r_solver_checks-20260823-084911
Final nonlinear crossover	paper/artifacts/enl_tanh_crossover-20260824-141207
Broad frozen-NTK pilot (failed)	paper/artifacts/enl_ntk_pilot-20260825
Restricted frozen-NTK factorial (passed)	paper/artifacts/enl_ntk_crossing_factorial-20260825
Exploratory nonlinear audit	results/enl_exact_regime_search-20260824-105003
Dense CDC ablation	results/e3_cdc_dense_ablation-20260823-090702

Reproduction commands. The ordinary runs can be regenerated from the repository root in the project environment:

```
python run_experiment.py e1 --config configs/e1_dense_rerun.yaml
python run_experiment.py e2 --config configs/e2_corrected_validation.yaml
python run_experiment.py e2r --config configs/e2r_solver_checks.yaml
python run_experiment.py enl --config configs/enl_tanh_crossover.yaml
python run_experiment.py e3 --config configs/e3_cdc_dense_ablation.yaml
```

The checked-in frozen evaluate configs are historical evidence records: they pin the original preflights and intentionally fail against later source. A new protocol run must first execute `enl-preflight`, then copy the matching evaluate config and explicitly repin `evaluation.preflight_dir` and `evaluation.manifest_sha256` before `enl-evaluate`. This creates a new study; it does not bitwise reproduce the archived dirty-source studies. The broad pilot pins manifest SHA-256 `f186d587...f0ffaf5` and prediction SHA-256 `9787d999...f9dfaa`; its evaluation verdict is failed. The restricted factorial pins manifest SHA-256 `bdcf02ab...a0db1` and prediction SHA-256 `c05b3181...6542`; its crossing-only verdict is passed. The exploratory E-NL command is deliberately omitted from the confirmatory list; its complete configuration and negative result remain archived in the repository.

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